Essays

|!|BCEEB0240000001|!| ★

Lung cancer is among the top 3 commonest cancer in Hong Kong in 2017. Describe how a malignant tumour may form in the lungs. Based on the risk factors for lung cancer, discuss how to prevent the development of lung cancer. (12 marks)

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Formation of a malignant tumour: (Max. 3)

- The genes that regulate cell division in lung cells undergo mutations. (1)
- The mutated lung cells divide uncontrollably and excessively, forming a mass of abnormal cells called a tumour. (1)
- Mutations further accummulate in the lung cells which become cancer cells (malignant). (1)
- The malignant tumour may spread to other parts of the body through the bloodstream or the lymphatic system, and invade other tissues to form new malignant tumours. (1)

Risk factors and preventive measures: (Max. 6)

Risk factor	Preventive measure
• Smoking / Secondhand smoke (1) Cigarette smoke contains many types of carcinogens which cause mutations in the cells lining the respiratory tract. (1)	• Smokers should quit smoking to reduce exposure to carcinogens in cigarette smoke. / Non-smokers should avoid breathing in the smoke of others. (1)
• Breathing in asbestos fibres (1) Asbestos is a carcinogen that has been used as an insulating material in buildings. When asbestos-containing material deteriorates, asbestos may be released to the air and people may inhale asbestos from the air. (1)	• Workers involved in the mining and processing of asbestos / People in old buildings under renovation should use proper protective equipment to minimize exposure to asbestos. (1) • People living in old buildings can hire experts to check asbestos levels in indoor air and remove any asbestos-containing materials. (1)
• Offspring who have inherited certain oncogenes have a higher risk of developing lung cancer. (1) • Cancer usually develops when mutations accumulate to a certain level. The risk of lung cancer gradually increases with age. (1)	• People with family history of lung cancer and elderly should regularly take screening tests for lung cancer. (1) Early detection of abnormal cells may help prevent further progression to cancer. (1)

Communication (Max. 3)

(12 marks)

Skills assessed	Marks
Understanding basic concepts	9
Communication	3

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Give an account on how obesity is linked to a high risk of cardiovascular diseases.

(12 marks)

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Obesity caused by improper diet and lack of physical activity: (Max. 3)

- When our energy intake is greater than our energy output, the excess energy will be stored as fat in our body and this causes weight gain. (1)
- When the body weight of a person is much greater than the normal range, the person is described as obese. (1)
- Obesity is usually caused by high-fat and/or high-sugar diet. (1)
- Lack of physical activity also leads to obesity. (1)

Relation between a high-fat diet and cardiovascular diseases: (Max. 4)

- Eating too many foods that are rich in saturated fats or cholesterol leads to high cholesterol content in the blood. (1)
- Excess cholesterol and other fatty substances in blood deposit on the inner walls of coronary arteries, forming plaque. (1)
- Plaque gradually increases in size, narrowing the lumen of the coronary artery. (1)
- A blood clot may form on the surface of plaque and this may completely block the lumen of the coronary artery. (1)
- When the coronary artery is blocked, the blood supply to the cardiac muscle becomes inadequate. (1)
- The cardiac muscle around the blocked artery cannot obtain enough oxygen and nutrients, and may die. (1)

Relation between insufficient physical activity and cardiovascular diseases: (Max. 2)

- When we are physically inactive, our energy output is low and we may become overweight or obese. (1)
- Obese people are more likely to have high blood pressure. (1)
- High blood pressure may cause damage to the delicate coronary arteries. (1)
- Being overweight or obese makes the heart work harder, increasing the risk of heart attack. (1)

Communication (Max. 3)

(12 marks)

Skills assessed	Marks
Understanding basic concepts	9
Communication	3

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